Yitzchak Shmalo

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Research interests

AI agents for mathematical and engineering research: a shared memory (the brain) with typed cards and learned edges, recall as a walk on a graph, reflexes compiled from recorded mistakes, a frontier layer that keeps agents off dead routes, gated write-back, and paired measurement of all of it. Mathematics of learning: random matrix theory for deep networks, pruning, stability and robustness, storage capacity, kernel methods. Flows on large graphs: money through crypto markets.

Appointments

2025 to date	Postdoctoral fellow, Einstein Institute of Mathematics, The Hebrew University of Jerusalem. Hosts: Sasha Sodin and Amit Daniely.
Summer 2024	Research intern, NASA Jet Propulsion Laboratory, California Institute of Technology, Pasadena. Uncertainty Quantification and Statistical Analysis group; mentors Otto Lamminpää and Jouni Susiluoto. Neural emulators of radiative transfer for imaging spectroscopy (EMIT, AVIRIS, OCO-2); low-rank and sparse kernel methods; random-matrix pruning. A later paper on the same emulation problem: arXiv:2609.00389 (2026).
2019 to 2025	Graduate teaching and research assistant, Department of Mathematics, Pennsylvania State University.

Education

2025	Ph.D. in Mathematics, Pennsylvania State University. Dissertation: *Applications of Random Matrix Theory to Deep Learning*. Advisor: Leonid Berlyand.
2017	M.A. in Mathematics, Yeshiva University. Advisor: Marian Gidea.
2016	B.A. in Mathematics, Yeshiva University.

Research with AI agents: the brain

Since 2025 I have directed AI agents in proving mathematics, in computational problems and on open problems, and in building the shared research memory they work from. The record of the work is more than a million Claude model calls and more than 17,000 Codex sessions, about half a trillion input tokens and more than 1.5 billion output tokens.

The math brain (github.com/yspennstate/math-brain-public): typed cards (definitions, theorems, lemmas, propositions, proof techniques, proofs, counterexamples, verified computations, open problems, dead routes) and typed links (depends on, proves, refutes, generalizes, specializes, analogous to, bridges to, cites, defines, supersedes). Private store 90,596 cards with 76,434 derived edges (28 Oct 2026); public snapshot 13,344 cards and 23,518 links (1 Nov 2026). It sits inside a larger private research brain of about 41,000 memories with 349,065 nodes and 15.5 million learned edges (1 Nov 2026), built from 15.3 million recorded events across 5,747 sessions.

Mechanisms. Recall by spreading activation over the graph, seeded by words and by meaning; a reflex layer that reads every prompt and action and surfaces recorded mistakes before they are repeated, with per-fire precision learned nightly; a frontier layer for open problems (64 problems, 2,623 routes, 2,197 claims on 28 Oct) that stops a worker on a route already on record; a gate that holds back any write that costs a held-out answer; a nightly consolidation pass; a mesh for concurrent agents with leases and

explicit handoffs; a speed programme that took a recall call from 26.8 s to 7.4 s and the prompt hook from 1.24 s to 0.20 s.

Measured, with paired exact tests. Three arms, same model, same questions, same time cap: *bare* (the model alone, no tools, no files), *repositories* (a shell and the checked-out repositories the memory was built from, no recall tool) and *memory* (the repositories arm plus recall over the store). 988 questions on research records, Claude agents: bare 18.9%, repositories 82.0%, memory 95.4%; on the pairs memory beat repositories 146 to 13 (p = 1.2e-29) and bare 757 to 1, in about half the time and at half the cost a right answer; on the 806 items both finished, memory and repositories were tied at 99.3% and 99.4%, so the lead is time. 2,000 new questions over 113 git repositories, Codex agents, memory (one recall a question) 93.7% against repositories 91.4% (83 to 43, p = 4.6e-4, n = 1,726), and 90.5% against 86.1% on an earlier bank (p = 1.0e-4, n = 967). A first 20-item test, memory 19 of 20 against bare 9 of 20 (p = 0.002). 1,000 recall items, memory 1,000 against repositories 538. 160 proof audits with planted faults, the right notes served against scrambled notes, 128 against 117 (p = 0.043, replicated). The repository also reports the tests that showed no gain (spreading one link out: same accuracy at 4.5 times the time) and the one the memory lost (a loaded machine, one recall taking 700 s, memory 39.9% against repositories 46.5%).

Results that came out of it (single author; AI systems did much of the mathematics under my direction, each paper with a declaration; audited by independent agent runs; not yet refereed): finite-state counterexamples to Ross's second queueing conjecture (arXiv:2609.25084); Vu's Conjecture 9.5 (false); Lyons and Peres Question 16.7 (false); Aldous and Fill Open Problem 7.30; Kosygina and Zerner Problem 5.3; the outstanding numerical conditions in the Ising perceptron storage-capacity theorem (arXiv:2607.11010); about 45 further repositories of the same kind at github.com/yspennstate.

Refereed journal articles

1. Y. Shmalo. Extreme least singular values of Gaussian row submatrices and a phase retrieval stability problem. *Bulletin of the Australian Mathematical Society*, published online 14 October 2026. doi:10.1017/S0004972726101713.
2. L. Berlyand, E. Sandier, Y. Shmalo, L. Zhang. Enhancing accuracy in deep learning using random matrix theory. *Journal of Machine Learning* 3 (2024), no. 4, 347-412. doi:10.4208/jml.231220.
3. Y. Shmalo. Stability of accuracy for the training of DNNs via the uniform doubling condition. *Annals of Mathematics and Artificial Intelligence* 92 (2024), no. 2, 439-483.
4. E. Prodan, Y. Shmalo. The K-theoretic bulk-boundary principle for dynamically patterned resonators. *Journal of Geometry and Physics* 135 (2019), 135-171.
5. M. Gidea, Y. Shmalo. Combinatorial approach to detection of fixed points, periodic orbits, and symbolic dynamics. *Discrete and Continuous Dynamical Systems* 38 (2018), no. 12, 6123-6148.

Preprints: deep learning, random matrices and kernels

6. L. Berlyand, T. Bourdais, H. Owhadi, Y. Shmalo. Pruning deep neural networks via the Marchenko-Pastur distribution. arXiv:2606.02608 (2026).
7. Y. Shmalo. The smallest eigenvalue of the neural tangent kernel of a deep tanh network in the ordered phase. 2026 (github.com/yspennstate/ntk-smallest-eigenvalue-ordered-phase). The exact constant and strict positivity of the smallest eigenvalue in the ordered phase, where earlier work had the order of magnitude only.
8. Y. Shmalo. Neural means and kernel corrections for operator learning. arXiv:2609.00389 (2026).
9. L. Berlyand, R. Sarapin, Y. Shmalo, V. Slavin, S. Sodin. Randomly initialized autoencoders: fixed points and edge-of-chaos. arXiv:2608.14638 (2026).

10. Y. Shmalo. A law of robustness for two-layer neural networks with arbitrary weights. arXiv:2607.07778 (2026).
11. Y. Shmalo. Mesoscopic rectangular spikes under subspace local laws: outlier values and singular subspaces. arXiv:2608.19594 (2026).
12. W. Krauth, Y. Shmalo. Path convergence under identical diffusion noise. Manuscript, 2026; the preprint has not yet been released. Convergence of finite-pattern diffusion paths to the infinite-pattern limit under a common Brownian driver, with a stable fluctuation theorem, endpoint tails and a bridge-type time change.
13. Y. Shmalo. The storage capacity of the Ising perceptron: verification of the outstanding numerical conditions. arXiv:2607.11010 (2026). Interval arithmetic, partly checked in Lean 4; most of the work by two AI systems under my supervision, as the paper states.
14. Y. Shmalo. Global stability of deep Gaussian ReLU networks. 2026.
15. Y. Shmalo. Math Brain: shared, auditable memory for multi-agent mathematical research. Full paper, October 2026; code and public snapshot at github.com/yspennstate/math-brain-public.
16. Y. Shmalo. Constructive adversarial fragility of DNNs, ResNets and ViTs. 2026.
17. Y. Shmalo. Balance, threshold calibration and adversarial neighbors on the sphere. 2026.
18. J. Jenkins, O. Krupchytskyi, Y. Shmalo. Deep learning weight pruning with RMT-SVD. arXiv:2303.08986 (2023).

Preprints: probability, random matrices and combinatorics

19. Y. Shmalo. Entropy and singularity of spectral and pivotal sets in planar percolation. arXiv:2609.15388 (2026).
20. Y. Shmalo. Fourier entropy of threshold functions and their compositions. arXiv:2610.16069 (2026).
21. Y. Shmalo. Extreme least singular values with redundancy, radial ensembles and phase retrieval. 2026.
22. Y. Shmalo. Mesoscopic additive spikes. 2026.
23. Y. Shmalo. Matrix Spencer for Toeplitz, Hankel and finite-entropy lift classes. 2026.
24. Y. Shmalo. Exact higher moments of random Toeplitz and Hankel limit laws; improved spectral averaging bound for the random Toeplitz law. 2026.
25. Y. Shmalo. Algorithm for computing representations of the braid group and Temperley-Lieb algebra. arXiv:2109.03306 (2021).
26. Y. Shmalo. Combinatorial proof of Kakutani's fixed point theorem. arXiv:1811.08454 (2018).
27. Y. Shmalo. A proof of Atanassov's conjecture and other generalizations of Sperner's lemma. arXiv:1708.07175 (2017).

Preprints: dynamics and topological physics (sole author, 2026)

Dense simultaneous distributional chaos; Strict moving horseshoes; Strict cubical horseshoes; Universal dense-open windows (Research Square, doi:10.21203/rs.3.rs-9222043/v1); Residual windows; Doubling and jamming (universal Chern Hamiltonians); Boundary and interface classes; A Roe-algebraic reduction. On ResearchGate and yspennstate.github.io.

Open-problem write-ups with AI agents, autumn 2026

Each a paper with its exact computations, a declaration of who did what, and an independent agent audit; not refereed. Ross's second queueing conjecture (arXiv:2609.25084); Vu's Conjecture 9.5 (github.com/yspennstate/gaussian-integer-spectra); Lyons and Peres Question 16.7 (cantor-harmonic-codings-singular); Aldous and Fill Open Problem 7.30 (arc-transitive-edge-heat); Kosygina and Zerner Problem 5.3

(positive-cookie-ballisticity); calibration along gradient descent in Gaussian tanh random-feature classifiers; exact power tails for a contractive nonlinear SGD model; sampling barriers for real Gaussian phase retrieval (31 October, in papers/ at math-brain-public); and about 40 more at github.com/yspennstate. Lean 4 components for probability results: github.com/yspennstate/probability-proof-components.

Software and data

math-brain-public: the memory, its tools, the starter that takes a team from an empty folder to a measured brain, and every test with its aggregates. ising-perceptron-capacity: certified interval computations. Public code for the Marchenko-Pastur pruning of vision transformers and CNNs. Private: the research brain and its hooks, the frontier layer, and a one-sweep tracer for all-hop flows on the Bitcoin transaction graph.

Awards

Teaching Excellence Recognition, Pennsylvania State University, 2022. Troxell Scholarship in Mathematics, Pennsylvania State University, 2020. Pettit Scholarship in Science, Pennsylvania State University, 2019. Teaching Excellence, Yeshiva University, 2017. Joseph Gunner Memorial Award for Excellence in Mathematics and Fisch Award, Yeshiva University, 2016.

Teaching

Pennsylvania State University, 2019 to 2025: undergraduate mathematics (Teaching Excellence Recognition, 2022). Yeshiva University, 2017 to 2019: graduate instructor, multivariable calculus, discrete structures and other courses (Teaching Excellence, 2017). Yeshiva Rabbi Samson Raphael Hirsch, 2017 to 2019: Algebra I and II, Geometry, curriculum design.

Skills

Python (NumPy, SciPy, PyTorch), Mathematica, MATLAB, R, LaTeX; Arb interval arithmetic; Lean 4 (with AI assistance); design and measurement of multi-agent AI systems (Claude, Codex).